Bag of Words

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Introduction to NLP | Bag of Words Model Watch this video

Corpus → Collection of documents

Review	class
loved it	1
liked it	1
Hated it	0

each row is a document

We need to convert these sentences into numbers.

BoW does that using/making a Vocabulary

Vocabular is made using the training data

- It is a set of words that we already know the meaning of

In this case, we will create a vocbulary by counting all the unique words in our corpus

Review	class
loved it	1
liked it	1
Hated it	0

Vocabulary

{loved, it, liked, hated}

FEATURE MATRIX

loved	it	liked	Hated
1	1	0	0
0	1	1	0
0	1	0	1

X

Target Vector

1
1
0

Y

Now, these are our embedded feature Per

doc. We have an X & Y & we can fit in any ML algo.

Preprocessing Techniques

- Lower case
- Remove all punctuations
- Stemming
  - Pulling out the word stems
    - Loving and Loves
    - If we take both, then number of features increase which causes our models robustness to decrease
    - Just take the STEM(not root) i.e. lov (NOT LOVE)
- Removing abbreviations and writing the full word
  - Wasn't -> Was not

- Bag of Words (BoW) is a technique that converts text into numbers so machine learning models can process it.
- Here is how it works:
- Vocabulary: It creates a master list of all unique words found across your entire dataset.
- Counting: It represents each document as an array (a vector) of numbers, showing how many times each vocabulary word appears in that specific text.
- Why "Bag"? It completely ignores word order, sentence structure, and grammar. It only cares about which words are present and how often, like throwing all the words into an unstructured bag.
- For example, if your vocabulary is ["I", "love", "dogs", "cats"], the sentence "I love dogs and I love cats" becomes the numerical array [2, 2, 1, 1] (ignoring "and" if it is a stop word).